L. Notifications, Communications & Templates

Standalone capability brief · Suggested module: **Communications Platform**

Purpose

Centralize reusable delivery mechanics, template lifecycle, preference handling, and delivery tracking for email, SMS, push, in-product messages, and webhooks.

1. Scope and capability areas

Template management

Version, review, localize, preview, and publish templates with clear ownership.

Channel preferences

Respect user and tenant delivery preferences and permitted channel choices.

Delivery and failover

Route through providers, handle provider outages, and support controlled failover.

Receipts and suppression

Track bounces, complaints, delivery receipts, unsubscribe state, and suppression lists.

Deduplication and tracking

Use stable notification identifiers and record lifecycle events without leaking sensitive content.

Webhook security

Sign outbound webhooks, retry safely, and protect replay and destination validation.

2. Shared contracts and standards

- Domain services submit a typed notification intent rather than choosing provider-specific details.
- Templates have versioned variables, localization behavior, and validation before publication.
- Consent, unsubscribe, and suppression rules are enforced centrally where legally and operationally appropriate.
- Delivery is asynchronous and has explicit accepted, queued, sent, delivered, failed, and suppressed states.

3. Recommended implementation approach

- Separate message intent from rendering and provider transport.
- Provide preview and test-send tools with safe non-production routing.
- Use bounded retry with provider-aware error classification and deduplication.
- Validate webhook destinations and signatures; provide replay-safe consumer guidance.
- Minimize sensitive data in templates, delivery logs, and provider payloads.

4. Failure behavior and operational expectations

- Provider failure should route to a permitted fallback or a retry queue without duplicating the user-visible message.

- Permanent failures such as invalid addresses should be suppressed or surfaced for correction rather than retried endlessly.
- Template rendering errors should be visible before send and should not fall back to leaking raw variables.

5. Suggested operational metrics

- Delivery success and latency by channel
- Bounce and complaint rates
- Provider failover frequency
- Template render failures
- Duplicate notification rate

6. Readiness and acceptance checklist

- ■ Domains can request notifications without provider-specific coupling.
- ■ Templates are versioned, localized, and validated before release.
- ■ Preferences and suppression rules are respected.
- ■ Webhook delivery is signed, retryable, and observable.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.